

● EASY

● ALGEBRA

Systems of two linear equations in two variables

02

10 questions · ~15 min

Q1

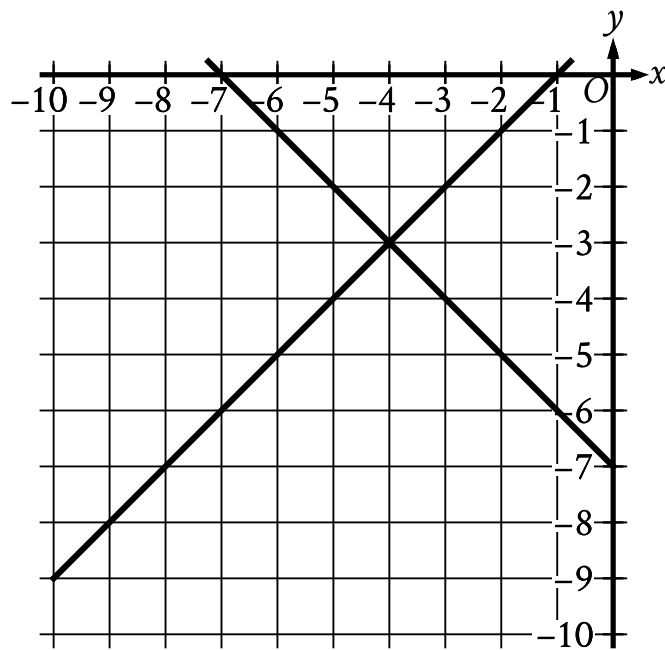
ALGEBRA

$$4x - 3y = 5$$

$$x = 8$$

What is the solution x, y to the given system of equations?

- A. 8, 9
- B. 8, -24
- C. 8, -9
- D. 8, 24



- For the first line in the system:
 - The line slants gradually down from left to right.
 - The line passes through the following points:
 - (negative 7 comma 0)
 - (negative 4 comma negative 3)
 - (0 comma negative 7)
- For the second line in the system:
 - The line slants gradually up from left to right.
 - The line passes through the following points:
 - (negative 5 comma negative 4)
 - (negative 4 comma negative 3)
 - (negative 3 comma negative 2)

The graph of a system of linear equations is shown. What is the solution x, y to the system?

A. 0, -7

B. 0, -3

C. -4, -3

D. -4, 0

Q3

ALGEBRA

$$5x = 15$$

$$-4x + y = -2$$

The solution to the given system of equations is x, y . What is the value of $x + y$?

A. -17

B. -13

C. 13

D. 17

A discount airline sells a certain number of tickets, x , for a flight for \$90 each. It sells the number of remaining tickets, y , for \$250 each. For a particular flight, the airline sold 120 tickets and collected a total of \$27,600 from the sale of those tickets. Which system of equations represents this relationship between x and y ?

- A. $\begin{cases} x + y = 120 \\ 90x + 250y = 27,600 \end{cases}$
- B. $\begin{cases} x + y = 120 \\ 90x + 250y = 120(27,600) \end{cases}$
- C. $\begin{cases} x + y = 27,600 \\ 90x + 250y = 120(27,600) \end{cases}$
- D. $\begin{cases} 90x = 250y \\ 120x + 120y = 27,600 \end{cases}$

Q5

ALGEBRA

$$\begin{aligned}y &= 2x + 3 \\ x &= 1\end{aligned}$$

What is the solution (x,y) to the given system of equations?

- A. $(1,2)$
- B. $(1,5)$
- C. $(2,3)$
- D. $(2,7)$

Q6

ALGEBRA

$$\begin{aligned}s + 7r &= 27 \\ r &= 3\end{aligned}$$

What is the solution r,s to the given system of equations?

- A. $(6,3)$
- B. $(3,6)$
- C. $(3,27)$
- D. $(27,3)$

$$4x = 20$$

$$-3x + y = -7$$

The solution to the given system of equations is x, y . What is the value of $x + y$?

- A. -27
- B. -13
- C. 13
- D. 27

Q8

GRID-IN ALGEBRA

$$x = 8$$

$$x + 3y = 26$$

The solution to the given system of equations is x, y . What is the value of y ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

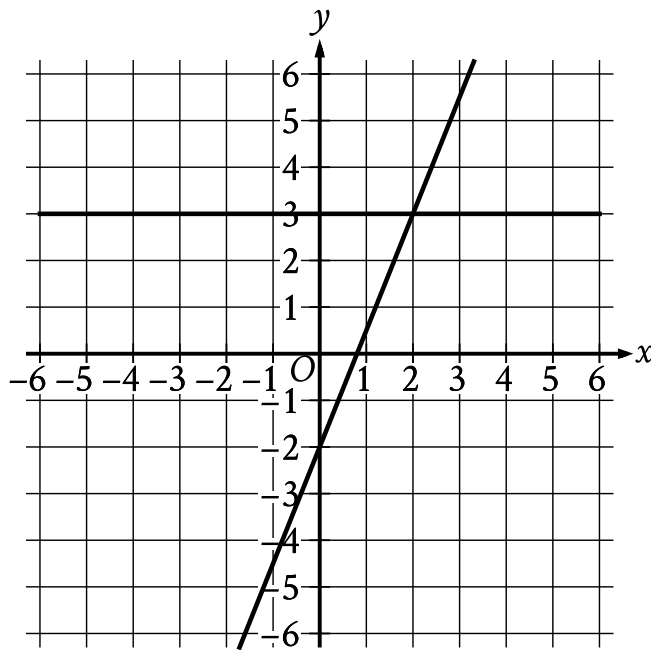
Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$x = 10$$

$$y = x + 21$$

The solution to the given system of equations is x, y . What is the value of y ?

- A. 2.1
- B. 10
- C. 21
- D. 31



- For the first line in the system:
 - The line is horizontal.
 - The line passes through the following points:
 - (negative 4 comma 3)
 - (0 comma 3)
 - (2 comma 3)
- For the second line in the system:
 - The line slants sharply up from left to right.
 - The line passes through the following points:
 - (negative 1 comma negative 4.5)
 - (0 comma negative 2)
 - (2 comma 3)

The graph of a system of linear equations is shown. What is the solution x, y to the system?

A. 0, 3

B. 1, 3

C. 2, 3

D. 3, 3